Globally-Mapped LiDAR Navigation for a Quadruped Robot

Executive Summary

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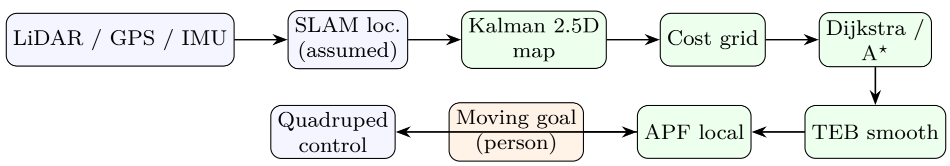
Objective. Reproduce and extend the classical navigation stack of a recent IEEE study on quadruped path planning [1] — *LiDAR mapping* → *localization* → *global planning* → *local obstacle avoidance* — and steer it toward our target application, autonomous **person-following** [2], where the robot tracks a moving person rather than a fixed goal. The whole stack was built from scratch in Python (`numpy` only; no heavy dependencies) and validated on synthetic, real automotive-LiDAR, and indoor RGB-D data.

What was built.

- **2.5D terrain map** — raw point clouds discretized into a per-cell height grid that keeps the steps/slopes/gaps a legged robot cares about, with a per-cell **Kalman filter** [3, 4] fusing repeated scans to reduce noise and flag low-confidence cells.
- **Global planner** — Dijkstra/A* shortest path [5, 6] over a traversability cost grid, then geometric (elastic-band / TEB-style) smoothing [7].
- **Local planner** — artificial potential field (Khatib) [8] for real-time obstacle avoidance, with a moving-goal mode for person-following.
- **Deployment path** — a ROS 2 / Nav2 node [9] that publishes the map as a costmap layer; plus 2D and interactive 3D visualizations.

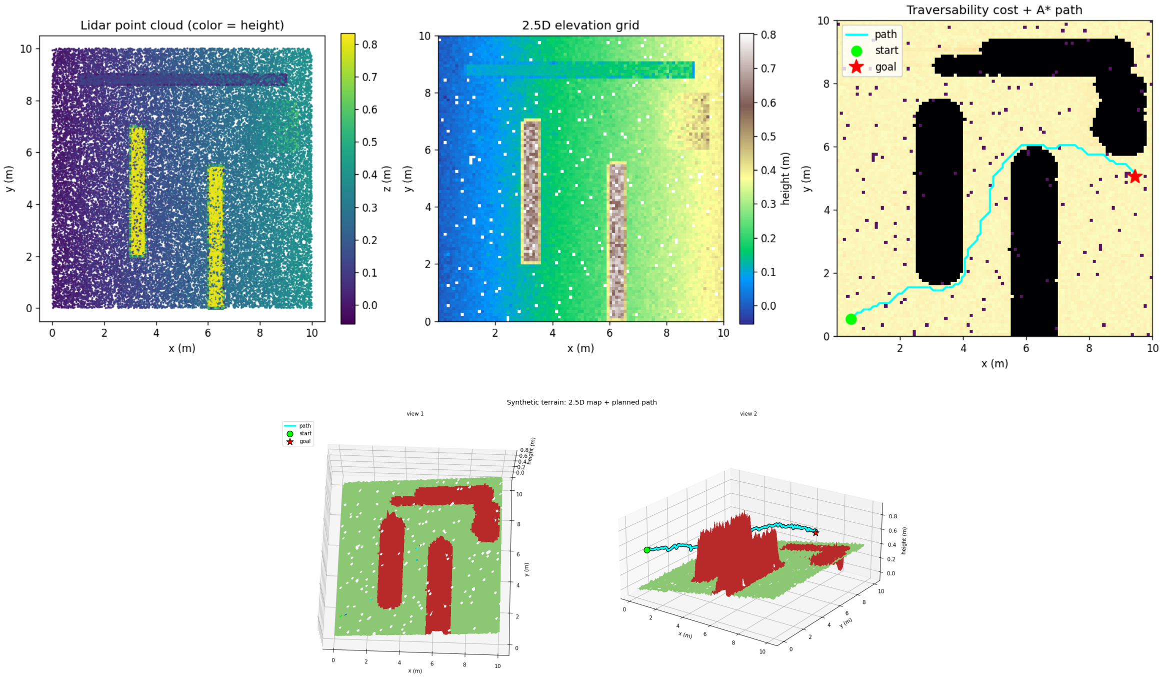


System architecture. Green = implemented; orange = person-following extension; white = assumed input.

Key findings.

1. **Clean LiDAR works out of the box; noisy RGB-D does not.** The same code handles real KITTI [10] automotive scans at default settings, but a single-view RGB-D floor has ∼15–20 cm of depth noise per cell and needs looser tolerances. *Traversability thresholds are sensor-specific, not universal constants.*
2. **Multi-scan fusion needs localization.** Fusing two real LiDAR scans taken from different robot poses inflated obstacle cells by ∼80% through ghosting; fusing a scan with itself did not. This empirically shows why the reference stack requires the SLAM/ICP [11] localization stage that we left as an assumed input.
3. **The global planner is essential, not decorative.** Pure APF toward a distant goal got trapped in a local minimum 4.4 m short; pairing it with the global path (APF following a moving sub-goal) reached the goal while keeping ∼0.5 m obstacle clearance. The person-following mode tracked a walking target at a ∼0.6 m mean gap.

Status & next steps. The mapping, planning, smoothing, avoidance, person-following, Nav2 integration, and visualization components are implemented and covered by 21 passing unit tests. The main remaining piece is the **localization** stage (ICP / LiDAR-inertial odometry) to supply robot poses for true multi-scan fusion, followed by the person-following **perception front-end** (camera-based detection and tracking) and on-robot testing through the Nav2 interface. A full technical report with methods, equations, and all figures accompanies this summary.



Example results. **Top:** point cloud → 2.5D elevation grid → cost grid with the planned A* path. **Bottom:** the same map in 3D, path draped over the terrain (green = traversable, red = obstacle).

References.

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